TEAM ASTRO

AeroSentinel

Satellite recovery and mission planning

We build the environment, tools and verifier.
Judges bring the agent.

9 task families / 3 levels / 6 tools

10-minute talk + 5 reference slides



The nine tasks

Task name	Level	What the agent must accomplish
1. EPS eclipse recovery	Easy	Repair power settings and retain eclipse battery reserve
2. ADCS detumble and pointing	Easy	Remove wheel bias and meet the pointing target
3. Thermal battery preservation	Medium	Repair heater control and keep battery temperature safe
4. RF downlink closure	Medium	Repair the radio configuration and link calculation
5. Safe-mode cascade	Medium	Recover two simultaneous subsystem faults safely
6. RF energy tradeoff	Hard	Deliver data within the contact and RF energy budget
7. Early pass recovery	Hard	Repair pointing and radio before the early pass closes
8. Observation window scheduling	Hard	Collect enough exposure within payload energy limits
9. Science store and forward	Hard	Schedule two acquisitions, finite storage and two downlinks

All nine use the same tools, seeded reset, baseline gates and programmatic verifier.

Guided tasks: constraints and difficulty

Guided task	Constraints the verifier enforces	What makes the repair non-trivial
EPS recovery	Fixed capacity; seeded battery reserve; voltage protection	Preserve energy through sunlight and eclipse
ADCS pointing	Protected pointing target and valid controls	Bias and controller response interact; deadband edits cannot lower the target
Thermal preservation	Trajectory temperature bounds and final ceiling	Safe final temperature cannot erase an earlier unsafe excursion
RF closure	Link margin, data delivery and script behavior	The calculation must work for live config and varied full-schema fixtures
Safe-mode cascade	Two seeded faults; all mandatory checks	Repair both without creating an irreversible safety failure

These are guided diagnostic tasks. The next four add deadlines, resource tradeoffs and multi-stage planning.

Hard tasks: constraints and planning challenge

Hard task	Decision and interacting constraints	Why final settings alone are insufficient
RF energy tradeoff	Choose rate and power within contact and RF energy budgets	High rate costs margin; high power costs energy
Early pass recovery	Repair pointing and radio before a short contact	A closed contact cannot be recovered later
Observation scheduling	Schedule useful exposure within payload energy	Out-of-window operation spends energy without exposure
Science store and forward	Two acquisitions and downlinks; finite buffer and two budgets	The first contact must free storage for acquisition two

Read exact contact times, targets, buffer and energy limits from docs/mission.json; values vary by seed.

Architecture and complete run workflow

Model API ↔ Harness: model loop, verifier and mission truth ↔ Sandbox: agent tools

Phase	What runs	Output / condition
1. Build + seeded reset	Build pinned images; create task(seed)	Same seed gives the same initial state hash
2. Reference and no-op	Reset separately for both baselines	Reference ≥0.95 and no-op ≤0.05
3. Gate + model reset	Reject failed gate; otherwise fresh reset	No model request if baselines fail
4. Model/tool loop	Validate call; execute sandbox tool	Updated protected state, telemetry, reward and errors
5. Final verification	Apply final programmatic checks	Reason, named checks, score and replayable action log

Only the harness reaches the model API. The agent sandbox has no outbound route.

What the agent sees and can do

Observations: docs/flight_manual.md, docs/mission.json, subsystem configs and current telemetry

Tool	Arguments	Effect
list_files	{}	List the onboard workspace
read_file	path; optional offset, limit ≤3500	Read a complete file in character ranges
write_file	path, content	Replace a configuration or Python file
run_python	code	Run a bounded check in the sandbox
step_orbit	minutes = 1..30	Advance the simulated mission and telemetry
finish	{}	Request final outcome verification

Persistent state: time, energy, temperature, pointing, exposure and science inventory. Ordinary attempts cost 30 simulated seconds.

Worked example: EPS recovery, seed 0

Read the manual and mission targets; inspect subsystems/eps.cfg and live telemetry.

Observed setting	Initial fault	Executed repair
solar_efficiency	0.41	0.72
undervoltage_threshold	2.80 V	3.20 V

1 Inspect files and diagnose the power deficit

2 Write the corrected config and test the changing world

3 Call finish; the verifier projects and checks the remaining orbit

Recorded result: 87.778% final battery reserve, all mandatory checks pass, reward 1.0 in 11 replies.

Worked example: science store and forward

Mission elapsed time	Executed / physical event	Verified effect
19.5 min; window 19-24 min	Enable science acquisition	270 s useful exposure; 513 kB generated
29-33 min	Downlink at contact one	513 kB delivered; buffer cleared
49-53 min	Collect in acquisition window two	240 s exposure; another 456 kB generated
74-78 min	Downlink at contact two	969 kB total delivery, zero loss
Across the 95-minute mission	Respect 598.5 kB storage and both energy budgets	Peak storage 513 kB; payload 1.00/1.08 Wh RF energy 1.541/1.626 Wh

Every window has its own quota. Constant duty needs ≥0.90 for exposure, but energy permits only ≤0.114 in seed 101.

Programmatic reward and final verification

Raw component	Standard	Observation	Science
Power / thermal	0.35 / 0.25	0.28 / 0.20	0.21 / 0.15
Pointing / communications	0.20 / 0.20	0.16 / 0.16	0.12 / 0.12
Exposure / delivered science	None	0.20 / none	0.20 / 0.20

reward = clamp((raw - initial no-op raw) / (1 - initial no-op raw), 0, 1)

Success ≥0.95 requires every mandatory check. Invalid configs score zero; catastrophic failures latch.

Science accounting: generated = buffered + delivered + lost. Final-file edits cannot recover missed exposure or lost data.

Isolation, errors and attempted shortcuts

Boundary / unexpected behavior	Implementation response	Measured evidence / limitation
Protected-state access	Separate verifier; authenticated controls UID 1001 agent / UID 1002 evaluator	12 live isolation checks Trusted Docker and host kernel
Fake telemetry / quotas	Grade protected state, not workspace copies	Forged telemetry reward 0.000; forged science inventory 0.521
Malformed reply / tool call	Bad reply counts; invalid tool schema never executes	Feedback, error log and no tool mutation
Runaway process / API failure	Bounded resources, descendant cleanup At most two transient retries	Retries share the 600-second episode
Missed window / data loss	Preserve irreversible mission history	Six scripted failure probes all score below success

Correctly scoring a model failure is expected. Undeserved success is the failure we guard against.

Reproduce a run from a fresh clone

Linux + Docker Compose

```
git clone https://github.com/devesh3001/ASTRO.git
cd ASTRO
cp .env.example .env
```

Edit .env for the organizer demo

```
MODEL=gpt-6-luna
API_KEY=<organizer key, kept private>
BASE_URL=https://litellm-hackathon.penpencil.guru
TASK_ID=eps_eclipse_recovery
SEED=0
MODE=full
APPEND_FILE=
```

Run once

```
docker compose up --build \
  --abort-on-container-exit \
  --exit-code-from harness
```

Expected: baseline gate, model loop, final score and a new logs/<run_id>/.

Offline smoke test: leave MODEL=mock/solver and credentials blank. All-task gates: MODE=baselines. Keep inputs/append.txt absent for Run A.

Measured results and preserved evidence

150 Linux tests / 12 live isolation checks / 9 reference-no-op gates passed

Genuine gpt-6-luna attempt	Outcome	Reward	Replies
EPS, seed 0, no append	success	1	11
RF energy, seed 101	success	1	11
Science, seed 101	success	1	28
Observation, seed 101	failure	0.8	12

474 / 474 stress expectations: 80 feasible plans succeed; 394 invalid choices are correctly rejected.

13 interface tests / 32 local HTTP requests are synthetic, not extra model runs. Four real attempts do not establish a general success rate.

Live Run A, Run B and the judge view

Run	Inputs judges select	What stays the same
A: named demo	EPS eclipse recovery, seed 0; no append	Same command, tools and verifier
B: unseen challenge	Judge-picked task and model; exact append file	Same interface, schemas and scoring No model-specific rules

Limits: build 180 s; baseline pair 120 s; 40 replies / 600 s; 10 tools per reply.
Tool operation 60 s; final verification has a separate 120 s.

Judge view: tools, reward, state hash and errors per reply.
Terminal reason: success / failure / step cap / time cap / crash.

10-minute talk and runs occur in parallel, then 6-minute Q&A. No intervention after a run starts.

Reference: exact task IDs

TASK_ID	Difficulty	Task objective
eps_eclipse_recovery	Easy	Recover eclipse energy reserve
adcs_detumble_pointing	Easy	Meet protected pointing target
thermal_battery_preservation	Medium	Recover safe temperature control
rf_downlink_closure	Medium	Repair radio link and calculation
safe_mode_cascade	Medium	Recover two subsystem faults
rf_energy_tradeoff	Hard	Meet contact delivery and RF energy
early_pass_recovery	Hard	Repair before early contact
observation_window_scheduling	Hard	Meet exposure and payload energy
science_store_forward	Hard	Two acquisitions, finite buffer, two contacts

Change TASK_ID and SEED in .env, then use the same Compose command.

Reference: files, logs and reproducibility

Where to look	Role / contents
harness/tasks.py; harness/science_mission.py	Seeded task generation, protected mission targets and plans
harness/tools.py; harness/runner.py	Six tool schemas, model loop, gates, budgets and telemetry
harness/env.py; sandbox/server.py	Environment control and sandbox boundary
logs/<run_id>/gate.json; manifest.json	Per-task reference/no-op results; source/build hashes
logs/<run_id>/episode*.jsonl	Inputs, raw replies, parsed actions, tool results, errors, state hash, reward, tokens, timings and final checks
logs/<run_id>/summary.json	Task outcome, termination, reward, reply count and duration
ASTRO.zip	Reference logs, trajectories, gates, design PDF and deck PDF

Same task + seed: same initial state. Same actions: same simulation. Model choices can differ.

Reference: limits, open questions and PS scope

Topic	What we can defend	What remains limited
Physics	Declared energy, scalar pointing, temperature, link and buffer model	No orbital propagation, 3-axis attitude or flight validation
Planning depth	Four coupled hard missions; independent feasible plans accepted	Five guided introductory tasks remain part of the mix
Robustness	Six negative probes, 12 live tests, bounded interface behavior	Finite tests do not prove every exploit impossible
Model portability	One common interface; synthetic protocol checks	One genuine deployment tested; unknown Run B untested
Reproduction	Pinned inputs, reset hashes, full action and outcome logs	Model sampling/provider reliability can differ
Submission	ASTRO naming; main branch; organizer invitations sent	Public repo conflicts with PS private rule Human submission / eligibility required

PS weighting: presentation 30, Run A 30, Run B 40. Honest scoring is the priority.

Solutions: the five guided recovery tasks

Task	Corrective action / seed-0 reference	Verify before finish
EPS eclipse recovery	solar_efficiency = 0.72 undervoltage_threshold = 3.20	Battery reserve and voltage safety
ADCS detumble / pointing	wheel_bias_rpm = 0; control_gain = 2.0 pointing_deadband_deg = 0.05	Protected pointing requirement
Thermal preservation	battery_heater_setpoint = 15.0 thermal_cutoff_temp = 45.0	Whole trajectory and final temperature
RF downlink closure	tx_power_dbm = 30.0; repair calc_margin() Read baud rate and path loss from comms.cfg	Live link margin, delivery and script checks
Safe-mode cascade	Repair both seeded faults. Seed 0: ADCS defaults plus tx_power_dbm = 30.0	All affected subsystem checks; no safety failure

Seed-0 reference values shown. Other seeds must meet their own mission targets; equivalent valid repairs are accepted.

Solutions: the four hard planning tasks

Hard task	Solution sequence	Required checks
RF energy tradeoff	Derive required rate from target / contact time; choose feasible baud rate and transmitter power. Seed 0: 38,400 bps, 32.0 dBm.	Delivered data, link margin and RF energy allocation
Early pass recovery	Repair pointing and radio before contact. Seed 0: control_gain 5.5; tx_power_dbm 31.4. Respect the protected maximum transmitter power.	Recover before pass closes; delivery and power constraints
Observation scheduling	Duty: 0 before window, 1 during, then 0. Account for 30-second ordinary tool costs. Meet useful exposure and payload energy.	Exposure quota and payload energy allocation
Science store and forward	Collect window 1; switch off; drain at contact 1. Collect window 2; switch off; drain at contact 2. Choose rate/power within the RF energy budget.	Every window quota; zero loss; delivery, buffer and both energy allocations

Derive timings and budgets from docs/mission.json for the active seed. Fixed schedules from another seed may fail.